Point-Source Neutrino Search: Analysis of NGC 1068 with IceCube Data

Master's Degree in Physics

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July 21, 2026

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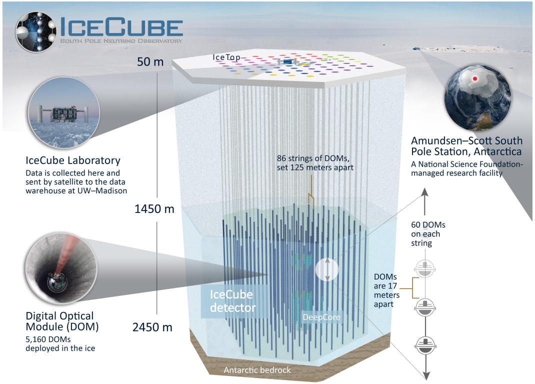
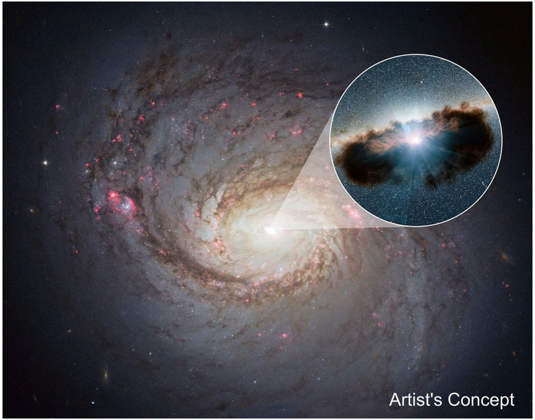
The IceCube Observatory and NGC 1068



Source: icecube.wisc.edu

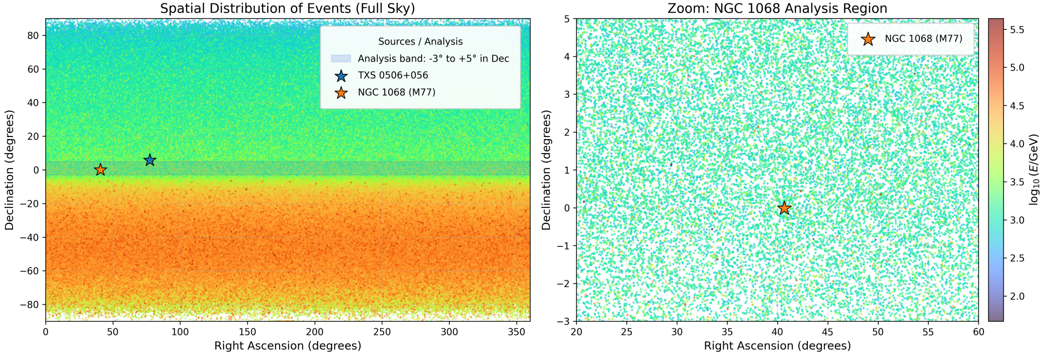


Source: science.nasa.gov

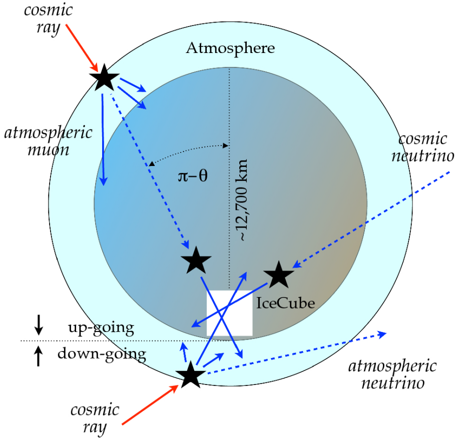
Dataset Selection & Spatial Cuts

The **Tracks dataset** provides the excellent angular resolution ($\sim 1^\circ - 2^\circ$) required for point-source astronomy.

The analysis is strictly confined to a narrow **declination band** ($\delta \in [-3^\circ, +5^\circ]$).



Data-Driven Background Estimation via RA Scrambling



Source: link.springer.com

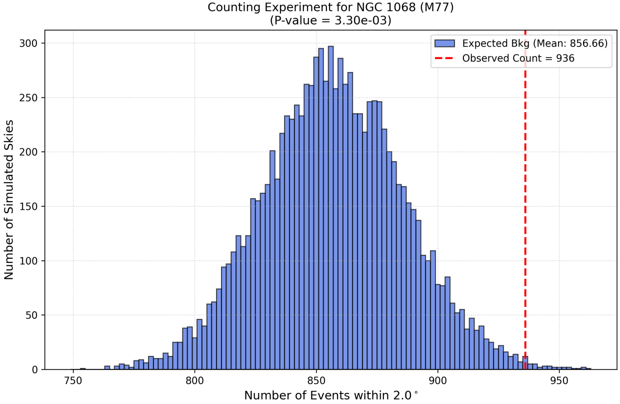
Analysis Region

- **Target:** NGC 1068 ($\delta \approx -0.01°$)
- **Band Cut:** $\delta \in [-3°, +5°]$
- Assumes locally uniform detector acceptance.

Background Modeling

- **Technique:** Right Ascension (RA) Scrambling
- **Process:** Fix δ_i, randomize α_i
- Data-driven baseline intrinsically accounting for exposure dependencies.

First signal search for NGC 1068



Counting Analysis Results

Parameter	**Value**
Simulated Skies	10^4
Observed Events	936
Exp. Background	856.7 ± 29.1
Excess Events	$+79.3$
***p*-value**	3.3×10^{-3}
Significance	2.72σ

Note: A simple binned circular search provides a solid baseline but does not exploit other important information.

Unbinned Spatial Likelihood

Mathematical Framework

$$\mathcal{L}(n_s) = \prod_{i=1}^{N} \left[\frac{n_s}{N} \mathcal{S}_i + \left(1 - \frac{n_s}{N}\right) \mathcal{B}_i \right]$$

- N: Total events in the declination band.
- n_s: Number of signal events (*free parameter*).

Probability Density Functions (PDFs)

Signal PDF ($\mathcal{S}_i$)

$$\mathcal{S}_i = \frac{1}{2\pi\sigma_i^2} \exp\left(-\frac{\psi_i^2}{2\sigma_i^2}\right)$$

Background PDF ($\mathcal{B}_i$)

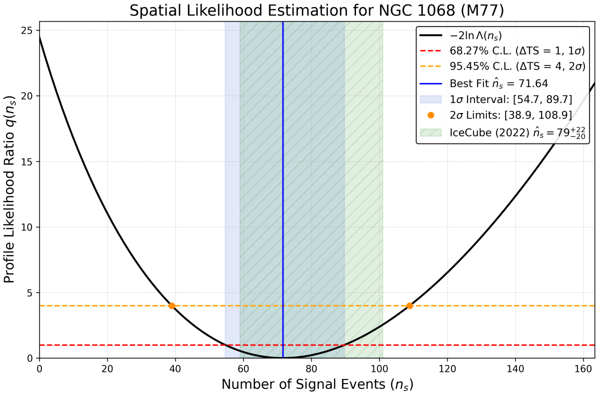
$$\mathcal{B}_i = \frac{1}{\Delta\Omega_{\text{band}}} = \frac{1}{2\pi(\sin\delta_{\text{max}} - \sin\delta_{\text{min}})}$$

Likelihood Scan and Best-Fit Results

- The Test Statistic (TS) is evaluated as the likelihood ratio against the null hypothesis:

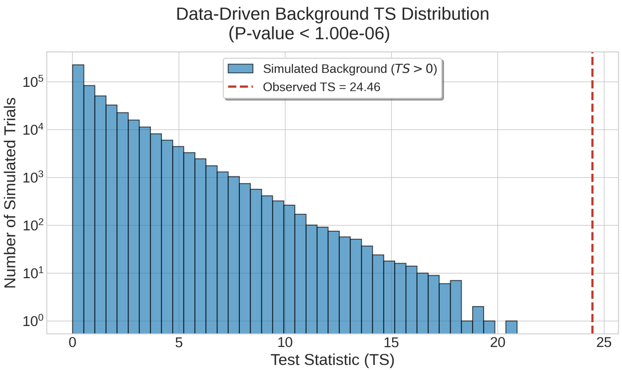
$$TS = 2\log\left(\frac{\mathcal{L}(\hat{n}_s)}{\mathcal{L}(n_s = 0)}\right)$$

- **Best-fit signal:** $\hat{n}_s = 71.64^{+18.08}_{-16.94} \quad 68.27\%$ C.L.
- **Observed TS:** $TS_{obs} = 24.46$
- **Significance:** $4.9\sigma \; p_{value} = 3.8 \times 10^{-7}$ (assuming the conditions of Wilks' theorem are satisfied)



Post-trial Significance via RA Scrambling

Empirical background TS distribution derived via Right Ascension scrambling ($N = 10^6$) to bypass Wilks' theorem asymptotic approximations at the target declination.



Statistical Results

- 0 background fluctuations out of 10^6 exceeded the observed TS.
- **p-value:** $< 1 \times 10^{-6}$

Conclusions & Outlook

A Solid Spatial-Only Baseline

- ✓ Successfully implemented simplified spatial-only likelihood analysis using IceCube Open Data.
- ✓ Identified a clear neutrino excess spatially consistent with NGC 1068.

Current Limitations & Future Directions

- ! **Energy Discrimination:** Neglecting lepton energy discards crucial separation power between hard astrophysical spectra and atmospheric backgrounds.
- ! **Computational Complexity:** True neutrino energy unfolding requires demanding advanced algorithms and Instrument Response Functions (IRFs).
- ! **Outlook:** Integrate energy-dependent PDFs to fully unlock the dataset's potential, and benchmark against the official IceCube Open Data **SkyLLH** framework.

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Thank you for listening!

Any questions?